

APMIC: Delivering hyper realistic AI assistants for a wide- range of scenarios

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ABOUT APMIC

By leveraging Google Cloud AI to develop and run natural language processing solutions, APMIC realizes more effective machine learning training and provides services with high availability and low operational effort.



About APMIC

Founded in 2017, APMIC is a Taiwanese tech company specializing in natural language processing technology. Its enterprise-grade machine intelligence solutions optimize interactions between human and machine, facilitating customer services, text analyses, and document reviews. APMIC currently provides services to 3.35 million users through 720 enterprise customers across various industries in Taiwan, Malaysia, Thailand, and Japan.

Industries: Technology

Google Cloud results

- Supports efficient machine learning training with higher convergence rate enabled by Vertex AI
- Shortens time required to set up a machine learning training environment from one week to one hour
- Helps reduce infrastructure operational workloads by 50 percent through GKE Autopilot
- Eliminates database failure and enhances data reading

Location: Taiwan

Saves
percentage
machine
learning
operational
costs

Run (<https://cloud.google.com/run/>), Cloud

Google Cloud (<https://cloud.google.com/>) compute
Engine (<https://cloud.google.com/compute/>),

Cloud Functions

(<https://cloud.google.com/functions/>)

Vertex AI (<https://cloud.google.com/vertex-ai/>)

Cloud Run (<https://cloud.google.com/run/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Compute Engine

(<https://cloud.google.com/compute/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Vertex AI (<https://cloud.google.com/vertex-ai/>)

**efficiency
with Cloud
SQL**

With the rapid development of large language models (LLMs), natural language processing solutions like chatbots and artificial intelligence (AI) text analyzers are becoming more widely used across industries. Leveraging the latest technologies, solution providers are now competing fiercely to offer the most powerful, easy-to-adopt AI services.

APMIC (<https://www.ap-mic.com/en>) has been advancing its natural language processing technology since 2017. Dedicated to optimizing interactions between human and machine, the Taiwanese tech company provides several enterprise-grade Model-as-a-Service (MaaS) solutions that facilitate business operations, including CaiGun, a custom chatbot service, and JustDig, an AI-powered text processing tool. Currently, APMIC offers services to 3.35 million users through 720 enterprise customers across various industries in Taiwan, Malaysia, Thailand, and Japan.

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quickly because
of Google's
focus on the
technology."

—Jerry Wu, Founder and
CEO, APMIC

"We've been improving our machine learning (ML)
models for years to help companies process text
more efficiently and provide better automated

customer services," says Jerry Wu, Founder and CEO at APMIC. "Our hyper realistic virtual assistants can be customized for various usage scenarios, and our customers are able to adopt them easily without having any programming or data organization skills."

Developing top-notch natural language processing solutions requires not only expertise but also powerful computing resources. However, the cloud platform that APMIC previously used to train its ML models didn't offer sufficient compute capabilities. To improve its products and ML training process, APMIC in 2019 decided to migrate its workloads and services to [Google Cloud](https://cloud.google.com/) (<https://cloud.google.com/>) for the advanced GPU and TPU resources available, as well as the leading role of Google in the AI domain.

"We need industry-leading compute capabilities to train our ML models efficiently and continue optimizing our virtual assistants," explains Wu. "Google Cloud provides powerful GPU and TPU chips that meet our needs, and its AI tools are advancing quickly because of Google's focus on the technology."

Enhancing ML training efficiency with Vertex AI

APMIC first adopted virtual machines (VMs) in [Compute Engine](https://cloud.google.com/compute) (<https://cloud.google.com/compute>) to build training environments for its ML models. Wu notes that setting up computing infrastructure for ML training was challenging on the cloud platform that the team previously used, because it had to

ensure the compatibility of the operating system, GPU toolkit, and software library, which all come with various versions. On top of that, the GPU chips available on the other cloud platform didn't possess enough memory for language model training, which dragged down the APMIC team's development process. Compute Engine provides the underlying infrastructure that are preset to match with different operating system versions, as well as advanced GPU chips like NVIDIA A100, accelerating APMIC's ML training.

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To further enhance its ML training efficiency, APMIC later moved its ML workloads to Vertex AI (<https://cloud.google.com/vertex-ai>), which is supported by the compute capabilities of Compute Engine and offers a complete ML training pipeline from data importing to inference with simple and well-guided steps. By automating parameter configurations, Vertex AI helps increase the convergence rate of APMIC's language models and reduce time required to run algorithms repeatedly. The trained models can also be easily shared with APMIC's enterprise customers for testing through an API in Vertex AI, which simplifies the development process.

"Vertex AI has significantly enhanced our ML training efficiency by providing excellent compute capabilities and streamlined training process," says Wu. "Before moving to Google Cloud, our engineer needed one week to set up an ML training environment. With Vertex AI, the whole setup process only takes an hour. Now, we only need one to two data engineers to train a large language model, which used to require a team of 10 engineers."

Supporting hyper realistic virtual assistants with diverse features

Apart from increasing ML training efficiency, leveraging Vertex AI has also enabled APMIC to improve its AI services easily. The Model Garden on Vertex AI offers a wide range of pre-trained models supporting features like optical character recognition (OCR) and automatic conversion between text and speech, allowing the APMIC to quickly meet its customers' diverse needs. By adopting PaLM 2 (<https://ai.google/discover/palm2/>), which according to Wu delivers responses one to two times faster with more elaborated content compared to other similar offerings, APMIC is able to build high-quality chatbots in multiple languages for its customers.

"We have high confidence in the AI capabilities of Google. Through Vertex AI, we can effortlessly take advantage of its latest technology advancement and provide AI services with top-grade quality, such as hyper realistic virtual assistants powered by PaLM 2," notes Wu.

Ensuring high service availability at half the cost

Another prominent advantage of using Google Cloud for APMIC is the access to fully managed and highly automated infrastructure tools. After the language models are fully trained in Vertex AI, the tech company deploys most of its natural language processing services in Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine>) (GKE) and

uses Cloud Run (<https://cloud.google.com/run>) and Cloud Functions (<https://cloud.google.com/functions>) to run API services with higher traffic. Since the resources of all three tools can be used according to the actual demand, APMIC is able to lower its operational costs by 50 percent compared to the time when it used VMs, which can only be launched by machine unit.

Employing GKE to run ML services also means higher availability, which is crucial to APMIC's business, because ML inference requires uninterrupted computing. Before moving to Google Cloud, the servers that APMIC used for deployment experienced downtime every several months. This meant that the team needed to spend time moving its services to new machines before the scheduled downtime to continue running ML inference and providing services smoothly. The migration has eliminated any downtime, and APMIC can also use

GKE Autopilot

(<https://cloud.google.com/kubernetes-engine/docs/concepts/autopilot-overview>)

to automate node management and scaling, which altogether leads to high service availability realized through 50 percent less workloads. Now, it only requires one engineer working half-time on the operations of all the ML projects of APMIC's 720 enterprise customers.

The high availability of APMIC's services is also enabled by Cloud SQL, which the tech company employs to store ML training and user data. Before the migration, APMIC's database could not support the high amount of data ingestion, resulting in

database failure once every two to three months. The high scalability of Cloud SQL has prevented such incidents, and by creating read replicas in Cloud SQL (<https://cloud.google.com/sql>), APMIC is able to offload read requests and enhance the cost-effectiveness of data reading.

Building easy-to-customize, more powerful language models

As generative AI is evolving rapidly, APMIC plans to broaden its natural language processing service range for more usage scenarios. It will enable its customers to train their own ML models on its platform to meet their specific needs, and provide virtual humans with custom appearance and voice. The tech company will also test the upcoming LLM of Google to explore new ways to upgrade its ML services.

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*—Jerry Wu, Founder and
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Wu says, "Natural language processing is a fast-growing and competitive field. Google Cloud has greatly facilitated our ML product development with its cutting-edge AI capabilities and high-performance cloud infrastructure. As its technology advances, we believe that we can continue enhancing the functionality and quality of our ML services."